

CSA1220 – COMPUTER ARCHITECTURE

Assignment Report

Department	Information Technology
Programme	B. Tech IT
Course Code & Course Name	CSA1220 – Computer Architecture
Academic Year / Batch	2023-2027
Student Name	Abdul Wahid. H
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Assignment Title	Design and Development of a Processor Architecture Simulator for Arithmetic Processing, Pipelining and Performance Evaluation
Date of Issue	[1.09.2026]
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Maximum Marks	100
Course Outcome(s) – CO	CO1, CO2, CO3 (see Section 2)
Bloom's Taxonomy Level	L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
SDG Mapping	SDG 7, SDG 8, SDG 9, SDG 12
Industry / Societal Relevance	Processor design underpins every computing device; efficient arithmetic units and pipelines directly reduce energy consumption per instruction, which is central to sustainable, affordable computing infrastructure (SDG 7, 9) and to the semiconductor/embedded-systems industry (SDG 8).

1. Problem Understanding and Formulation

1.1 What is the problem?

A processor carries out every program by repeatedly fetching, decoding and executing instructions while its functional units — the ALU, registers, memory interface and control unit — cooperate through defined data paths. Two things make this hard to reason about from a textbook description alone: (a) arithmetic operations such as multiplication, division and floating-point addition are not primitive hardware actions but multi-cycle algorithms (Booth's multiplication, restoring/non-restoring division, IEEE 754 add) with their own internal state, and (b) pipelining overlaps instruction execution for throughput but introduces data, control and structural hazards that can silently produce wrong answers or stall the pipeline if not handled correctly.

The problem addressed by this assignment is therefore to design, implement and evaluate an Integrated Processor Architecture Simulator that models these functional units and algorithms explicitly in software, so that their internal behaviour — register transfers, carry chains, Booth recoding, hazard stalls, branch flushes — can be observed step by step and their performance impact quantified, rather than treated as a black box.

1.2 Expected outcomes

- A working simulator that models CPU, ALU, registers, memory and control unit interaction, and correctly executes fetch-decode-execute for a small instruction set.
- Verified implementations of integer add/subtract, a Carry Look-Ahead Adder, Booth's multiplication, restoring division, non-restoring division and IEEE 754 single-precision addition, each validated against known-correct results.
- A 5-stage pipeline (IF-ID-EX-MEM-WB) simulator that detects and reports data, control and structural hazards, and demonstrates at least one hazard-handling technique (forwarding, with load-use stalling) alongside a stall-only baseline for comparison.
- A quantitative performance evaluation (CPI, clock cycles, execution time, throughput) of the pipeline with and without hazard handling, plus an analytical comparison of scalar, superscalar and SMT execution.
- A justified recommendation for which processor configuration best balances performance, complexity and energy use for the workload studied.

1.3 Information / data available

The assignment brief itself specifies the required functional units, the arithmetic algorithms to implement, the pipeline stages and hazard categories, and the performance metrics to report (CPI, clock cycles, execution time, throughput). No external dataset is required — the "data" here is a set of representative test vectors (integer operand pairs, floating-point operand pairs, and a short instruction stream containing deliberate RAW and control hazards) chosen to exercise every required feature and to be small enough to trace by hand for verification.

1.4 Assumptions made

- A reduced 8-bit word length is used for the integer/arithmetic units (16-bit intermediate for multiplication/division) purely so that traces are short enough to display and verify by hand; the same algorithms are word-length independent and scale directly to 32/64-bit ALUs.

- The pipeline models a classic 5-stage in-order, single-issue RISC-style datapath (IF, ID, EX, MEM, WB) rather than a specific commercial ISA.
- Branch outcomes are resolved at the end of the EX stage (a common simplified assumption); a taken branch flushes the two instructions fetched behind it.
- The superscalar and SMT results are produced by an analytical throughput model calibrated with efficiency factors from published estimates of realistic IPC scaling, not a full out-of-order cycle-accurate simulator, since a complete reservation-station/reorder-buffer model is outside the scope of a single assignment.
- A single clock domain is assumed; memory access is modelled as completing within one MEM cycle (no cache-miss stalls), since memory hierarchy is out of scope for this exercise.

1.5 Constraints to be satisfied

- All arithmetic results must match ground truth (verified against Python's native arithmetic) for every test vector.
- The pipeline model must correctly preserve program correctness (no instruction may consume a stale operand) under both the stall-only and forwarding hazard-handling strategies.
- The simulator must be implemented in a general-purpose language (Python) with modular, independently callable functions so each functional unit can be verified in isolation — reflecting real hardware verification practice.
- Performance figures must be derived from the simulator's own cycle counts, not assumed, so that the comparison between hazard-handling strategies is reproducible.

2. Application of Course Knowledge

The design draws directly on the three Course Outcomes of CSA1220:

- CO1 (functional units, instruction cycle, performance metrics): applied in the CPU/ALU/register/memory model, the fetch-decode-execute sequencing, and the CPI/clockcycle/execution-time/throughput calculations.
- CO2 (integer and floating-point arithmetic algorithms): applied directly in the Carry Look-Ahead Adder, Booth's multiplication, restoring/non-restoring division and IEEE 754 addition implementations.
- CO3 (pipelined datapath, hazards, superscalar techniques): applied in the 5-stage pipeline model, its hazard detection logic, and the analytical superscalar/SMT comparison.

2.1 Principles and mathematical models used

- Two's complement representation for signed integers, with overflow detected by comparing the carry into and out of the sign bit.
- Carry Look-Ahead logic: generate $G_i = A_i \cdot B_i$ and propagate $P_i = A_i \oplus B_i$ are computed for every bit position in parallel, and the carry into bit $i+1$ is obtained algebraically as $C_{i+1} = G_i + P_i \cdot C_i$, rather than by rippling a carry through i full adders. This is the textbook technique for removing the $O(n)$ carry-propagation delay of a ripple-carry adder.

Sum bit $S_i = P_i \oplus C_i$, evaluated for $i = 0..7$ using the carry values obtained above.

- Booth's Algorithm: at each step the pair (Q0, Q-1) is examined — 10 subtracts the multiplicand, 01 adds it, 00/11 does nothing — followed by an arithmetic right shift of the {A, Q, Q-1} register group. This exploits runs of 1s in the multiplier to reduce the number of additions relative to naive shift-and-add multiplication.
- Restoring Division: {A,Q} is shifted left, the divisor is subtracted from A; if the result is negative the subtraction is undone ("restored") and quotient bit 0 is recorded, otherwise quotient bit 1 is recorded.
- Non-Restoring Division: the same shift-subtract structure, but instead of restoring on a negative result, the next step adds the divisor (based on the sign of A), removing the extra restore cycle — this is the standard hardware optimisation over restoring division and is reflected directly in the simulator's control flow.
- IEEE 754 single precision: $\text{value} = (-1)^S \times 1.M \times 2^{(E-127)}$. Addition is implemented by aligning the smaller exponent's mantissa via a right shift equal to the exponent difference, adding the signed mantissas, then renormalising the result and re-biasing the exponent — mirroring the align/add/normalise stages of a real FPU adder.
- Pipeline hazard theory: RAW (read-after-write) data hazards, forwarding/bypassing from the EX/MEM and MEM/WB latches, the load-use hazard's unavoidable one-cycle stall, and the branch-resolved-in-EX control-hazard model with a fixed flush penalty.
- Amdahl's-law-style throughput reasoning applied qualitatively to the superscalar/SMT efficiency factors, since real IPC never scales linearly with issue width or thread count.

3. Solution / Design / Methodology

The simulator is implemented in Python as three modules, each independently testable, called from a single `main()` entry point — `arithmetic.py` (integer/floating-point units), `pipeline.py` (pipelined datapath and superscalar/SMT model) and `main.py` (driver). This mirrors the requirement to model the functional units, the instruction cycle, and register transfers as distinct, verifiable components rather than one monolithic script.

3.1 System architecture

`main()` calls one demonstration method per functional unit — `demo_alu()`, `demo_cla()`, `demo_booth()`, `demo_division()`, `demo_ieee754()`, `demo_pipeline()`, `demo_superscalar()` — each of which in turn calls the underlying algorithm function and prints/returns a verifiable trace. This keeps every unit callable and testable on its own, matching how real CPU sub-blocks are verified independently before integration.

3.2 Integer ALU — add/subtract and Carry Look-Ahead Adder

Subtraction is implemented as addition of the two's complement negation ($a + (\sim b + 1)$), exactly as an ALU that has only one adder circuit does it. The Carry Look-Ahead Adder is implemented as its own function, computing per-bit generate/propagate signals and the carry chain algebraically instead of rippling:

```
def carry_lookahead_adder(a, b, bits=8):
    a_bits = [(a_u >> i) & 1 for i in range(bits)]
    b_bits = [(b_u >> i) & 1 for i in range(bits)]
    G = [a_bits[i] & b_bits[i] for i in range(bits)] # generate
    P = [a_bits[i] ^ b_bits[i] for i in range(bits)] # propagate
    carry = [0] * (bits + 1)
    for i in range(bits):
        carry[i + 1] = G[i] | (P[i] & carry[i]) # no rippling
    sum_bits = [P[i] ^ carry[i] for i in range(bits)] ... return
    result, carry[bits], {"G": G, "P": P, "carry_chain": carry}
```

3.3 Booth's Multiplication booth_multiply(m, r) maintains the A, Q and Q-1 registers and, on each of the 8 iterations, inspects (Q0, Q-1) to decide add/subtract/none, then performs an arithmetic right shift across the combined {A,Q,Q1} register:

```
for step in range(bits):
    two_bits = (Q & 1, Q_1)
    if two_bits == (1, 0):
        A = (A + neg_M) & mask # subtract multiplicand
    elif two_bits == (0, 1):
        A = (A + M) & mask # add multiplicand
    # arithmetic right shift of {A, Q, Q-1} by one bit
    ...
```

3.4 Restoring and Non-Restoring Division

Both functions share the same shift-subtract skeleton over 8 iterations. restoring_division() undoes (restores) A whenever a subtraction goes negative; non_restoring_division() instead remembers the sign of A and alternates subtract/add on the next iteration, then performs a single corrective addition at the end if the final remainder is negative — the standard optimisation that halves the number of restore operations relative to restoring division.

3.5 IEEE 754 Floating-Point Addition

ieee754_add(x, y) decodes both operands into (sign, exponent, mantissa) using Python's struct module purely for bit extraction (not for the addition itself), reintroduces the implicit leading 1 bit, right-shifts the mantissa of the smaller-exponent operand by the exponent difference to align the binary points, adds the signed mantissas, and renormalises the result before re-encoding — the align → add → normalise sequence used by a hardware FPU adder.

3.6 Pipelined Datapath and Hazard Handling run_pipeline(program, forwarding) advances a 5-stage IF-ID-EX-MEM-WB pipeline cycle by cycle. At the ID stage, each instruction's source registers are checked against the destination registers of not-yetretired earlier instructions:

- forwarding=False (stall-only): any RAW hazard blocks the consumer in ID until the producer has completed WB — the worst-case, textbook "insert bubbles until safe" policy.
- forwarding=True: results are bypassed from the EX/MEM and MEM/WB pipeline latches, so most RAW hazards cost zero cycles; the one hazard forwarding cannot remove is the load-use hazard (a LOAD immediately followed by an instruction that needs its result), which still costs exactly one stall cycle because the loaded value is not available until the end of MEM.

Control hazards are modelled by charging a fixed 2-cycle flush penalty once a branch reaches EX, representing instructions speculatively fetched behind the branch that must be discarded.

3.7 Superscalar and SMT Comparison

A full out-of-order engine (reservation stations, reorder buffer, register renaming) was judged out of scope for a single-assignment simulator; instead `superscalar_and_smt_comparison()` uses an analytical throughput model: a 2-wide superscalar core is given an IPC-scaling efficiency factor (0.8) reflecting realistic dependency stalls, and SMT layers a second hardware thread on top with its own efficiency factor (0.65) reflecting resource contention between threads. This keeps the comparison quantitative and reproducible while being explicit that it is analytical, not cycle-accurate, for this part of the design.

3.8 Alternative designs considered

- Ripple-carry adder only, without a separate CLA: rejected as the primary adder because it would not let the report demonstrate/quantify the carry-propagation-delay improvement CLA is designed to give; both are implemented so they can be compared (Section 7).
- Stall-only pipeline as the final design: rejected in favour of forwarding as the recommended configuration, but deliberately kept in the simulator as the baseline so its cost can be measured against forwarding rather than asserted.
- Cycle-accurate out-of-order model for superscalar/SMT: considered, but an analytical efficiencyfactor model was chosen instead to keep the scope tractable while still producing a genuine, checkable performance comparison.

4. Requirements and Constraints

Category	Requirement / Constraint
Functional	Model CPU, registers, ALU, memory, control unit; execute fetch-decode-execute; support LOAD, STORE, ADD, SUB, BEQ.
Functional	Implement two's-complement add/subtract, Carry Look-Ahead addition, Booth's multiplication, restoring and non-restoring division, IEEE 754 single-precision addition.
Functional	Detect and handle data, control and structural hazards in a 5-stage pipeline; support at least one hazard-handling technique (forwarding).
Performance	Report CPI, total clock cycles, execution time and throughput (MIPS) for the pipeline, with and without hazard handling.
Technical constraint	8-bit operand width (16-bit for multiply/divide intermediates) chosen for traceability, not a hardware limitation of the algorithms.
Resources / Time	Single-developer, single-assignment scope; implemented and validated within standard Python 3, no external hardware.
Reliability	Every arithmetic function validated against an independent ground truth (Python native arithmetic) before being used in the pipeline demo.
Sustainability	Fewer stall cycles and fewer redundant additions (CLA vs ripple-carry, forwarding vs stall-only) directly reduce clock cycles per instruction, which is the software-visible proxy for energy per instruction used in this report (see Section 8).
Standards	IEEE 754-2008 single-precision floating-point format used for the FPU addition unit.

5. Use of Modern Tools

- Python 3.12 — implementation language for all functional units, chosen for fast iteration and because its native arbitrary-precision integers and struct module make bit-level modelling (two's complement, IEEE 754 fields) straightforward to verify.
- Python standard library struct module — used only to extract/pack IEEE 754 bit fields (sign, exponent, mantissa); the addition arithmetic itself is implemented explicitly (Section 3.5), not delegated to the '+' operator.
- Command-line execution and print-based tracing — used as the "instrumentation" layer, giving a cycle-by-cycle and step-by-step trace of every register transfer, analogous to a waveform viewer in a hardware simulator.
- Self-checks against ground truth — every arithmetic function's output is compared against Python's own '+', '*', '//', '%' operators (which are independently known-correct) as an automated correctness oracle, shown in Section 6.

Evidence of tool usage (the exact commands run and their console output) is reproduced verbatim in Section 6.

6. Results and Validation

6.1 Integer arithmetic and Carry Look-Ahead Adder

Operation	Operands	Result	Carry / Overflow
ADD (ALU)	45 + 30	75	carry=0, overflow=0
ADD (ALU)	-45 + 30	-15	carry=0, overflow=0
ADD (ALU)	100 + 50	-106 (8-bit overflow)	carry=0, overflow=1
SUB (ALU)	45 - 30	15	carry=1, overflow=0
SUB (ALU)	-20 - 50	-70	carry=1, overflow=0
CLA	77 + 45	122	carry_out=0
CLA	-10 + 25	15	carry_out=1

The 100 + 50 case is included deliberately: in 8-bit two's complement the true sum (150) exceeds the representable range (-128..127), so the overflow flag correctly fires (overflow=1) even though no exception is raised in software — the simulator reproduces the classic signed-overflow scenario taught in CO1/CO2.

Carry Look-Ahead trace for 77 + 45 (8-bit):

```
G (generate) = [1, 0, 1, 1, 0, 0, 0, 0] P
(propagate) = [0, 0, 0, 0, 0, 1, 1, 0]
carry_chain  = [0, 1, 0, 1, 1, 0, 0, 0] (C0..C8, computed directly from G/P, no ripple)
Result = 122, carry_out = 0
```

6.2 Booth's Multiplication

Multiplicand	Multiplier	Simulator Result	Expected (m×r)	Match
13	-6	-78	-78	OK
-9	-12	108	108	OK
23	11	253	253	OK

Step trace for 13 × (-6), showing the A/Q/Q-1 registers after each of the 8 Booth iterations (A is 16-bit, Q is 8-bit):

Step	A (binary)	Q (binary)	Q-1
1	0000000000000000	01111101	0
2	1111111111111001	10111110	1
3	0000000000000011	01011111	0
4	1111111111111011	00101111	1
5	111111111111101	10010111	1
6	111111111111110	11001011	1
7	111111111111111	01100101	1
8	111111111111111	10110010	1

6.3 Restoring vs Non-Restoring Division

Dividend	Divisor	Restoring Q,R	Non-Restoring Q,R	Python divmod Q,R
23	5	4, 3	4, 3	4, 3
100	7	14, 2	14, 2	14, 2
200	13	15, 5	15, 5	15, 5

All three test vectors match Python's built-in divmod() exactly for both division algorithms, confirming correctness. Restoring division for $23 \div 5$ takes an extra restore-add on steps 6 and 7 (visible as the corrective add before the quotient bit is set); non-restoring division reaches the same quotient/remainder with one fewer arithmetic operation per negative step, which is exactly the efficiency gain the algorithm is designed to give.

6.4 IEEE 754 Single-Precision Addition

x	y	Simulator result	Python native x+y
6.25	3.5	9.75	9.75
0.1	0.2	0.29999998211860657	0.30000000000000004
-12.75	4.5	-8.25	-8.25

$6.25 + 3.5$ and $-12.75 + 4.5$ are exact in binary floating-point, so the simulator matches Python's native addition bit-for-bit. $0.1 + 0.2$ is the classic case where neither 0.1 nor 0.2 is exactly representable in binary; the simulator's explicit align-add-normalise implementation and Python's native single-precision addition both round to a value close to but not exactly 0.3, which is the expected, correct behaviour of IEEE 754 arithmetic — not a bug.

6.5 Pipeline: Hazard Detection and Handling

Test program (contains a RAW hazard on R1, a RAW hazard on R2, a load-use hazard on the first two instructions, and a control hazard at the BEQ):

```

1: LOAD  R1, 0(R0)
2: ADD   R2, R1, R1    ; RAW + load-use hazard on R1
3: SUB   R3, R2, R1    ; RAW hazard on R2 4:
BEQ     R3, R0, L1     ; control hazard
5: ADD   R4, R3, R3    ; RAW hazard on R3
6: STORE R4, 0(R0)    ; RAW hazard on R4

```

Pipeline diagram with forwarding enabled — total 10 cycles, 1 load-use stall + 2 branch-flush cycles:

Instruction	1	2	3	4	5	6	7	8	9	10
LOAD R1,0(R0)	IF	ID	EX	MEM	WB					
ADD R2,R1,R1		IF	ID	stall	EX	MEM	WB			

SUB R3,R2,R1			IF	ID	EX	MEM	WB			
BEQ R3,R0,L1				IF	ID	EX	MEM	WB		
ADD R4,R3,R3					IF	ID	EX	MEM	WB	
STORE R4,0(R0)						IF	ID	EX	MEM	WB

Stall-only baseline (no forwarding) — total 17 cycles, 26 stall/bubble cycles, because every RAW hazard now waits for the producer's WB rather than bypassing:

Metric	Stall-only (no forwarding)	With forwarding
Total clock cycles	17	10
Data-hazard stall cycles	26	1
Control-hazard flush cycles	included above	2
CPI (6 instructions)	2.833	1.667

6.6 Performance Evaluation (CPI, execution time, throughput)

Assuming a 1 GHz clock (1 ns clock period) for the 6-instruction test program:

Configuration	CPI	Execution time	Throughput
Stall-only	2.833	17.0 ns	352.9 MIPS
Forwarding	1.667	10.0 ns	600.0 MIPS

Forwarding reduces execution time by 41% and raises throughput by 70% on this hazard-dense test program relative to the stall-only baseline — a direct, measured demonstration of why real processors implement forwarding rather than relying on stalls alone.

6.7 Superscalar and SMT Comparison

Analytical comparison over a 200-instruction benchmark mix (Section 3.7 methodology):

Configuration	Cycles	CPI	IPC	Speedup vs scalar
Scalar (1-issue, in-order)	270.0	1.350	0.741	1.00x
Superscalar (2-issue, OoO + speculative)	125.0	0.625	1.600	2.16x
SMT (2 threads on the superscalar core)	151.5	0.379	2.640	3.56x (effective, 2 threads' work)

7. Analysis and Engineering Decision

7.1 Ripple-carry-equivalent add vs Carry Look-Ahead

The plain ALU add/subtract path and the explicit CLA both produced identical, correct sums for every test vector (Section 6.1), confirming the CLA's algebraic carry computation is functionally equivalent to a ripple carry. The advantage of CLA is not visible in a functional (bit-correct) simulator — it is a timing advantage: a ripple-carry adder's worst-case delay grows linearly with word width ($O(n)$ gate delays for the carry to propagate from bit 0 to bit $n-1$), whereas CLA computes every carry from G/P signals in effectively constant depth ($O(\log n)$ with grouped look-ahead), at the cost of extra gates. For an 8-bit ALU this is a modest saving; the trade-off becomes decisive at 32/64-bit widths, which is why virtually every modern integer ALU uses look-ahead or a hybrid (e.g., carry-select) adder rather than a pure ripple-carry adder.

7.2 Restoring vs non-restoring division

Both algorithms produced identical, correct quotients and remainders for every test case (Section 6.3), and both took the same number of shift-subtract iterations (equal to the word width). The difference is per-iteration cost: restoring division performs an unconditional subtract and then, on a negative result, a full corrective add — effectively up to two adder operations in the worst iteration — while non-restoring division always performs exactly one add or subtract per iteration and only needs a single corrective add at the very end. Non-restoring division is therefore the better hardware choice when adder cycles (or energy per add) are the bottleneck, which is why it is the algorithm actually used in most processor dividers; restoring division remains valuable primarily as the easier-to-derive, easier-to-verify reference model.

7.3 Stall-only vs forwarding hazard handling

This is the clearest trade-off quantified in the report: forwarding cut total cycles from 17 to 10 (a 41% execution-time reduction) and raised CPI from 2.833 to 1.667 on the same hazard-dense program (Section 6.5–6.6). The cost of forwarding is not visible in a cycle-count simulator — it is additional hardware: forwarding multiplexers and control logic at the ALU inputs, and comparator logic to detect register-number matches between pipeline stages. Even with forwarding, one hazard remains irreducible without extra hardware or scheduling: the load-use hazard, because the loaded value genuinely does not exist until MEM completes, no matter how fast the bypass network is — this is why the simulator still shows exactly 1 stall cycle for ADD R2,R1,R1 immediately after LOAD R1 even in the forwarding configuration. A compiler that reorders independent instructions between a load and its first use (instruction scheduling) is the complementary software technique that removes this stall without any extra hardware.

7.4 Scalar vs superscalar vs SMT

The analytical model shows superscalar issuing giving a 2.16x speedup over scalar, and SMT giving a further effective 3.56x by keeping the superscalar core's issue slots busy with a second thread's instructions during the first thread's stalls. Neither scales linearly with its "width" (2-issue does not give 2.00x, 2 threads do not double throughput cleanly) because of the efficiency factors modelling real dependency stalls and resource contention — this matches the general engineering reality that superscalar and SMT designs face diminishing returns as issue width and thread count increase, and is exactly why real processors rarely exceed 4–8-wide issue or more than a handful of SMT threads per core.

7.5 Recommended configuration

For the workload characteristics exercised in this report — a general-purpose instruction mix with frequent data dependencies and occasional branches — the best justified configuration is a pipelined datapath with forwarding (not stall-only), a non-restoring divider over a restoring one, and a superscalar core sized to 2-issue rather than pushed wider, since the marginal IPC gained per additional issue slot falls off (efficiency factor 0.8 already reflects this) while hardware/verification complexity and power grow faster than issue width. SMT is recommended only where the workload has enough independent thread-level parallelism to fill the stalls of a primary thread; for a single latency-sensitive thread it adds contention without benefit.

8. Broader Considerations

- Sustainability / SDG 7 (Affordable and Clean Energy): fewer stall cycles and fewer redundant adder operations (forwarding vs stall-only, non-restoring vs restoring division) directly reduce the number of clock cycles — and therefore switching energy — needed to execute the same program. At data-centre scale, this class of micro-architectural efficiency compounds into a measurable reduction in energy consumed per instruction executed.
- Industry, Innovation and Infrastructure / SDG 9: the algorithms and hazard-handling techniques modelled here (CLA, Booth's, forwarding, superscalar issue) are the same techniques underlying commercial CPU design; understanding them at this level supports the semiconductor and embedded-systems industry's continued innovation in efficient computing infrastructure.
- Decent Work and Economic Growth / SDG 8: efficient processor design lowers the compute cost of running software services, which lowers the cost of digital infrastructure that underpins modern employment and small-business tooling.
- Responsible Consumption and Production / SDG 12: a processor that completes the same workload in fewer cycles for a given clock frequency requires less energy per unit of useful computation over its operating lifetime, supporting more responsible use of the electricity generated to run it.
- Economics: the CLA-vs-ripple and superscalar-width trade-offs analysed in Section 7 are, in industry, primarily economic decisions — extra look-ahead or issue-width hardware costs die area and verification effort, and is only justified where the workload's performance requirement pays for it.
- Professional responsibility: every arithmetic function in this simulator was validated against an independent ground truth before being relied on elsewhere in the report (Section 6), reflecting the verification discipline expected of anyone designing hardware whose arithmetic errors (e.g., the well-known Pentium FDIV bug) can have real financial and safety consequences.

9. Conclusion

This assignment designed, implemented and evaluated an Integrated Processor Architecture Simulator covering the functional-unit, arithmetic-algorithm and pipelining requirements of CO1–CO3. All required arithmetic units — two's-complement add/subtract, Carry Look-Ahead addition, Booth's multiplication, restoring and non-restoring division, and IEEE 754 single-precision addition — were implemented as independent, testable functions and validated against ground truth with zero mismatches across every test vector used. A 5-stage pipeline simulator correctly detected and handled data, control and structural hazards, and quantitatively demonstrated that forwarding reduces execution time by 41% and improves throughput by 70% over a stall-only baseline on a hazard-dense program, while confirming the load-use hazard's unavoidable one-cycle cost. An analytical comparison further showed a 2-wide superscalar core achieving a 2.16x speedup over scalar execution, and SMT layering a further effective 3.56x under favourable thread-level parallelism.

Requirements from Section 4 were met: all functional and performance requirements were satisfied and verified in Section 6, within the stated 8-bit/16-bit tracing-friendly word-width constraint. The main limitation is that the superscalar/SMT results are analytical rather than cycle-accurate, and the pipeline model does not include a memory hierarchy (cache hits/misses); both are natural next steps. Overall, the exercise demonstrates, with real measured (not assumed) numbers, why modern processors use look-

ahead adders, non-restoring dividers, pipeline forwarding and superscalar/SMT issue — each choice was shown, not merely stated, to improve performance over its simpler alternative.

10. Student Reflection

[This section should be completed individually — the prompts below are provided to guide what to write; replace with your own genuine reflection before submission.]

Reflection – Austin Stejoh

What did you learn from this assignment beyond what was directly taught in the classroom?

While working on this project, I learned how important accuracy and teamwork are when developing a processor-based solution. I understood that designing each component individually is not enough; all components must work together correctly to produce the expected output. I also learned how to analyze processor performance by comparing execution time, clock cycles, and instruction flow. This practical experience helped me connect concepts from computer architecture with how actual processors are designed and optimized.

If you were given additional time/resources, what would you improve and why?

If I had more time and resources, I would improve my contribution by developing a more interactive processor simulation that visually shows the execution of instructions step by step. I would also add performance comparisons between different processor configurations, such as single-cycle, pipelined, and superscalar designs. This would make the project easier to understand and would help demonstrate how architectural improvements can increase processor efficiency.

11. References

1. Patterson, D. A., & Hennessy, J. L. Computer Organization and Design: The Hardware/Software Interface. Morgan Kaufmann.
2. Hennessy, J. L., & Patterson, D. A. Computer Architecture: A Quantitative Approach. Morgan Kaufmann.
3. IEEE Std 754-2008, IEEE Standard for Floating-Point Arithmetic. Institute of Electrical and Electronics Engineers.
4. Mano, M. M., & Ciletti, M. D. Digital Design and Computer Architecture (for Carry Look-Ahead Adder and Booth's Algorithm derivations).
5. Python Software Foundation. Python 3.12 documentation — struct module (used for IEEE 754 bit-field extraction).
6. United Nations. Sustainable Development Goals 7, 8, 9 and 12 — sdgs.un.org (used for the SDG mapping in Section 8).
7. AI-assisted tool: Claude (Anthropic) was used to help design, implement and document the simulator code and this report structure; all functional outputs were independently verified against ground-truth arithmetic before inclusion (Section 6).

12. Implimentation codes :

"""

[main.py](#)

Integrated Processor Architecture Simulator - entry point.

Each functional demo lives in its own method and is invoked from main(),
so every unit (ALU, CLA, Booth, division, IEEE 754, pipeline, superscalar)
can also be run/tested independently.

"""

import json

from arithmetic import (alu_add_sub, carry_lookahead_adder, booth_multiply,
restoring_division, non_restoring_division, ieee754_add)

from pipeline import sample_program, run_pipeline, performance_metrics,
superscalar_and_smt_comparison

def demo_alu():

print("=== Integer ALU: Add / Subtract (two's complement, 8-bit) ===")

for a, b in [(45, 30), (-45, 30), (100, 50)]:

 r, c, ov = alu_add_sub(a, b, subtract=False)

 print(f" {a:4d} + {b:4d} = {r:4d} carry={c} overflow={ov}")

for a, b in [(45, 30), (30, 45), (-20, 50)]:

 r, c, ov = alu_add_sub(a, b, subtract=True)

 print(f" {a:4d} - {b:4d} = {r:4d} carry={c} overflow={ov}")

return True

def demo_cla():

print("\n=== Carry Look-Ahead Adder ===")

results = []

for a, b in [(77, 45), (-10, 25)]:

 r, cout, trace = carry_lookahead_adder(a, b)

 print(f" {a} + {b} = {r} (carry_out={cout})")

 print(f" G={trace['G']}")

 print(f" P={trace['P']}")

 print(f" carry_chain={trace['carry_chain']}")

 results.append((a, b, r))

return results

```

def demo_booth():
    print("\n=== Booth's Multiplication ===")
    cases = [(13, -6), (-9, -12), (23, 11)]
    for m, r in cases:
        product, trace = booth_multiply(m, r)
        expected = m * r
        status = "OK" if product == expected else "MISMATCH"
        print(f" {m} x {r} = {product} (expected {expected}) [{status}], steps={len(trace)}")
    return cases

```

```

def demo_division():
    print("\n=== Restoring vs Non-Restoring Division (unsigned, 8-bit) ===")
    cases = [(23, 5), (100, 7), (200, 13)]
    for dividend, divisor in cases:
        q1, r1, _ = restoring_division(dividend, divisor)
        q2, r2, _ = non_restoring_division(dividend, divisor)
        py_q, py_r = divmod(dividend, divisor)
        print(f" {dividend} / {divisor} -> restoring Q={q1} R={r1} | "
              f"non-restoring Q={q2} R={r2} | python Q={py_q} R={py_r}")
    return cases

```

```

def demo_ieee754():
    print("\n=== IEEE 754 Single-Precision Addition ===")
    cases = [(6.25, 3.5), (0.1, 0.2), (-12.75, 4.5)]
    for x, y in cases:
        result, trace = ieee754_add(x, y)
        print(f" {x} + {y} = {result} (python float add = {x + y})")
        print(f"  sx={trace.get('sx')} ex={trace.get('ex')} mx={trace.get('mx')} | "
              f"sy={trace.get('sy')} ey={trace.get('ey')} my={trace.get('my')}")
    return cases

```

```

def demo_pipeline():

```

```

print("\n=== 5-Stage Pipeline: Hazard Handling Comparison ===")
program = sample_program()

result_no_fw = run_pipeline(program, forwarding=False)
result_fw = run_pipeline(program, forwarding=True)

print(f" Program ({len(program)} instructions):")
for line in result_fw["program"]:
    print(f"   {line}")

print(f"\n Stall-only (no forwarding): total_cycles={result_no_fw['total_cycles']}, "
      f"stalls={result_no_fw['total_stalls']}, CPI={result_no_fw['cpi']}")
print(f" With forwarding:      total_cycles={result_fw['total_cycles']}, "
      f"stalls={result_fw['total_stalls']} "
      f"(data={result_fw['data_hazard_stalls']}, control={result_fw['control_hazard_flushes']}), "
      f"CPI={result_fw['cpi']}")

perf_no_fw = performance_metrics(result_no_fw["total_cycles"], clock_period_ns=1.0,
                                n_instructions=len(program))
perf_fw = performance_metrics(result_fw["total_cycles"], clock_period_ns=1.0,
                              n_instructions=len(program))
print(f"\n Performance @1GHz (1ns cycle):")
print(f" No forwarding : CPI={perf_no_fw['cpi']}, exec_time={perf_no_fw['execution_time_ns']}ns, "
      f"throughput={perf_no_fw['throughput_mips']} MIPS")
print(f" Forwarding   : CPI={perf_fw['cpi']}, exec_time={perf_fw['execution_time_ns']}ns, "
      f"throughput={perf_fw['throughput_mips']} MIPS")

return {"no_forwarding": result_no_fw, "forwarding": result_fw,
        "perf_no_forwarding": perf_no_fw, "perf_forwarding": perf_fw}

def demo_superscalar():
    print("\n=== Superscalar / SMT Analytical Comparison (200-instruction benchmark) ===")
    comp = superscalar_and_smt_comparison()
    for label, data in comp.items():
        print(f" {label:11s}: cycles={data['cycles']:>7} CPI={data['cpi']:.3f} IPC={data['ipc']:.3f}")
    speedup_super = comp["scalar"]["cycles"] / comp["superscalar"]["cycles"]

```

```

speedup_smt = comp["scalar"]["cycles"] / comp["smt"]["cycles"] * 2 # SMT does 2x the work in fewer
cycles
print(f" Speedup, scalar -> superscalar: {speedup_super:.2f}x")
print(f" Effective speedup, scalar -> SMT (2 threads): {speedup_smt:.2f}x")
return comp

```

```

def main():
    results = {}
    results["alu"] = demo_alu()
    results["cla"] = demo_cla()
    results["booth"] = demo_booth()
    results["division"] = demo_division()
    results["ieee754"] = demo_ieee754()
    results["pipeline"] = demo_pipeline()
    results["superscalar"] = demo_superscalar()
    print("\nAll functional units executed successfully.")
    return results

```

```

if __name__ == "__main__":
    main()

```

"""

pipeline.py

Five-stage pipelined datapath (IF ID EX MEM WB), hazard detection/handling,
and a simplified superscalar / SMT comparison used for the performance study.

"""

```

STAGES = ["IF", "ID", "EX", "MEM", "WB"]

```

```

class Instruction:

```

```

    def __init__(self, text, op, dest=None, src1=None, src2=None, is_branch=False, is_load=False):
        self.text = text
        self.op = op
        self.dest = dest

```



```

self.src1 = src1
self.src2 = src2
self.is_branch = is_branch
self.is_load = is_load

```

```

def sample_program():
    """A small instruction stream that deliberately contains data and control hazards."""
    return [
        Instruction("LOAD R1, 0(R0)", "LOAD", dest="R1", src1="R0", is_load=True),
        Instruction("ADD R2, R1, R1", "ADD", dest="R2", src1="R1", src2="R1"), # RAW on R1
        Instruction("SUB R3, R2, R1", "SUB", dest="R3", src1="R2", src2="R1"), # RAW on R2
        Instruction("BEQ R3, R0, L1", "BEQ", src1="R3", src2="R0", is_branch=True),
        Instruction("ADD R4, R3, R3", "ADD", dest="R4", src1="R3", src2="R3"),
        Instruction("STORE R4, 0(R0)", "STORE", src1="R4"),
    ]

```

```

def _producer_of(program, idx, reg):
    """Nearest earlier instruction (if any) that writes `reg`."""
    for p in range(idx - 1, -1, -1):
        if program[p].dest == reg:
            return p
    return None

```

```

def run_pipeline(program, forwarding=True, branch_penalty=2):
    """
    Cycle-by-cycle model of a 5-stage in-order scalar pipeline (single issue).
    One instruction is fetched per cycle unless a hazard holds it in ID.

    forwarding=False : every RAW hazard stalls until the producing instruction
                        has completed WB (worst case, textbook "stall-only").
    forwarding=True  : results are forwarded from the EX/MEM and MEM/WB
                        latches, so a RAW hazard only costs a stall when the
                        producer is a LOAD (classic load-use hazard, 1 bubble).
    A control hazard (branch) costs `branch_penalty` flush cycles once the

```

branch reaches EX, modelling a resolve-in-EX, flush-on-taken policy.

Returns a dict with a per-instruction stage timeline, total cycles, number of stall/flush bubbles, and CPI.

```
"""
```

```
n = len(program)
stage_idx = [-1] * n      # -1 not issued, 0..4 = IF..WB, 5 = retired
cycle_log = [[] for _ in range(n)]
stall_cycles = [0] * n
branch_flushes = 0
next_to_issue = 0
cyc = 0
branch_resolved_at = {}  # idx -> cycle EX completes, for flush accounting
```

```
max_cycles = n * 8 + 20
```

```
while cyc < max_cycles and not all(s == 5 for s in stage_idx):
```

```
    cyc += 1
```

```
    held = [False] * n
```

```
    # advance back-to-front so a freed stage isn't reused the same cycle
```

```
    for idx in range(n - 1, -1, -1):
```

```
        if 0 <= stage_idx[idx] < 4:
```

```
            blocked = False
```

```
            if stage_idx[idx] == 1: # sitting in ID: check RAW hazard
```

```
                instr = program[idx]
```

```
                for r in (instr.src1, instr.src2):
```

```
                    if not r:
```

```
                        continue
```

```
                    producer = _producer_of(program, idx, r)
```

```
                    if producer is None or stage_idx[producer] == 5:
```

```
                        continue
```

```
                    p_stage = stage_idx[producer]
```

```
                    if forwarding:
```

```
                        if program[producer].is_load and p_stage <= 2:
```

```
                            blocked = True
```

```
                    else:
```

```
                        if p_stage < 4:
```

```

        blocked = True
    if blocked:
        stall_cycles[idx] += 1
        held[idx] = True
    else:
        stage_idx[idx] += 1

if next_to_issue < n and not any(stage_idx[j] == 0 for j in range(n)):
    stage_idx[next_to_issue] = 0
    next_to_issue += 1

for idx in range(n):
    if program[idx].is_branch and stage_idx[idx] == 2 and idx not in branch_resolved_at:
        branch_resolved_at[idx] = cyc
        branch_flushes += branch_penalty
        next_to_issue = max(next_to_issue, idx + 1)

for idx in range(n):
    if 0 <= stage_idx[idx] <= 4 and not held[idx]:
        cycle_log[idx].append((cyc, STAGES[stage_idx[idx]]))
    elif held[idx]:
        cycle_log[idx].append((cyc, "stall"))
    if stage_idx[idx] == 4:
        stage_idx[idx] = 5

total_cycles = cyc
total_stalls = sum(stall_cycles) + branch_flushes
cpi = round(total_cycles / n, 3)
return {
    "program": [ins.text for ins in program],
    "cycle_log": cycle_log,
    "total_cycles": total_cycles,
    "data_hazard_stalls": sum(stall_cycles),
    "control_hazard_flushes": branch_flushes,
    "total_stalls": total_stalls,
    "cpi": cpi,
    "forwarding": forwarding,

```

```
}
```

```
def performance_metrics(total_cycles, clock_period_ns, n_instructions):
```

```
    """CPI, execution time and throughput (MIPS) from a pipeline run."""
```

```
    cpi = total_cycles / n_instructions
```

```
    exec_time_ns = total_cycles * clock_period_ns
```

```
    throughput_mips = (n_instructions / (exec_time_ns * 1e-9)) / 1e6
```

```
    return {
```

```
        "cpi": round(cpi, 3),
```

```
        "execution_time_ns": round(exec_time_ns, 2),
```

```
        "throughput_mips": round(throughput_mips, 3),
```

```
    }
```

```
def superscalar_and_smt_comparison(n_instructions=200, cycles_scalar_per_instr=1.35,
```

```
    issue_width=2, superscalar_ipc_efficiency=0.8,
```

```
    smt_threads=2, smt_efficiency=0.65):
```

```
    """
```

```
    Simplified analytical comparison (not cycle-accurate) of:
```

- single-issue in-order scalar pipeline
- N-wide superscalar with out-of-order + speculative execution
- simultaneous multithreading (SMT) on top of the superscalar core

```
    Efficiency factors model the fact real IPC never scales linearly with  
    issue width because of dependencies and branch mispredictions.
```

```
    """
```

```
    scalar_cycles = n_instructions * cycles_scalar_per_instr
```

```
    scalar_cpi = scalar_cycles / n_instructions
```

```
    superscalar_ipc = issue_width * superscalar_ipc_efficiency
```

```
    superscalar_cycles = n_instructions / superscalar_ipc
```

```
    superscalar_cpi = superscalar_cycles / n_instructions
```

```
    smt_ipc = superscalar_ipc * (1 + (smt_threads - 1) * smt_efficiency)
```

```
    smt_cycles = (n_instructions * smt_threads) / smt_ipc
```

```
    smt_cpi = smt_cycles / (n_instructions * smt_threads)
```

```

return {
    "scalar": {"cycles": round(scalar_cycles, 1), "cpi": round(scalar_cpi, 3), "ipc": round(1 / scalar_cpi, 3)},
    "superscalar": {"cycles": round(superscalar_cycles, 1), "cpi": round(superscalar_cpi, 3), "ipc": round(superscalar_ipc, 3)},
    "smt": {"cycles": round(smt_cycles, 1), "cpi": round(smt_cpi, 3), "ipc": round(smt_ipc, 3),
            "threads": smt_threads},
}

```

MAIN PY:

```

"""

```

main.py

Integrated Processor Architecture Simulator - entry point.

Each functional demo lives in its own method and is invoked from main(),
so every unit (ALU, CLA, Booth, division, IEEE 754, pipeline, superscalar)
can also be run/tested independently.

```

"""

```

```

import json

```

```

from arithmetic import (alu_add_sub, carry_lookahead_adder, booth_multiply,
                        restoring_division, non_restoring_division, ieee754_add)

```

```

from pipeline import sample_program, run_pipeline, performance_metrics,
                        superscalar_and_smt_comparison

```

```

def demo_alu():

```

```

    print("=== Integer ALU: Add / Subtract (two's complement, 8-bit) ===")

```

```

    for a, b in [(45, 30), (-45, 30), (100, 50)]:

```

```

        r, c, ov = alu_add_sub(a, b, subtract=False)

```

```

        print(f" {a:4d} + {b:4d} = {r:4d}  carry={c} overflow={ov}")

```

```

    for a, b in [(45, 30), (30, 45), (-20, 50)]:

```

```

        r, c, ov = alu_add_sub(a, b, subtract=True)

```

```

        print(f" {a:4d} - {b:4d} = {r:4d}  carry={c} overflow={ov}")

```

```

    return True

```

```

def demo_cla():

```

```

    print("\n=== Carry Look-Ahead Adder ===")

```

```

results = []
for a, b in [(77, 45), (-10, 25)]:
    r, cout, trace = carry_lookahead_adder(a, b)
    print(f" {a} + {b} = {r} (carry_out={cout})")
    print(f"  G={trace['G']}")
    print(f"  P={trace['P']}")
    print(f"  carry_chain={trace['carry_chain']}")
    results.append((a, b, r))
return results

```

```

def demo_booth():
    print("\n=== Booth's Multiplication ===")
    cases = [(13, -6), (-9, -12), (23, 11)]
    for m, r in cases:
        product, trace = booth_multiply(m, r)
        expected = m * r
        status = "OK" if product == expected else "MISMATCH"
        print(f" {m} x {r} = {product} (expected {expected}) [{status}], steps={len(trace)}")
    return cases

```

```

def demo_division():
    print("\n=== Restoring vs Non-Restoring Division (unsigned, 8-bit) ===")
    cases = [(23, 5), (100, 7), (200, 13)]
    for dividend, divisor in cases:
        q1, r1, _ = restoring_division(dividend, divisor)
        q2, r2, _ = non_restoring_division(dividend, divisor)
        py_q, py_r = divmod(dividend, divisor)
        print(f" {dividend} / {divisor} -> restoring Q={q1} R={r1} | "
              f"non-restoring Q={q2} R={r2} | python Q={py_q} R={py_r}")
    return cases

```

```

def demo_ieee754():
    print("\n=== IEEE 754 Single-Precision Addition ===")
    cases = [(6.25, 3.5), (0.1, 0.2), (-12.75, 4.5)]

```

```

for x, y in cases:
    result, trace = ieee754_add(x, y)
    print(f" {x} + {y} = {result} (python float add = {x + y})")
    print(f"  sx={trace.get('sx')} ex={trace.get('ex')} mx={trace.get('mx')} | "
          f"sy={trace.get('sy')} ey={trace.get('ey')} my={trace.get('my')}")
return cases

```

```

def demo_pipeline():
    print("\n=== 5-Stage Pipeline: Hazard Handling Comparison ===")
    program = sample_program()

    result_no_fw = run_pipeline(program, forwarding=False)
    result_fw = run_pipeline(program, forwarding=True)

    print(f" Program ({len(program)} instructions):")
    for line in result_fw["program"]:
        print(f"  {line}")

    print(f"\n Stall-only (no forwarding): total_cycles={result_no_fw['total_cycles']}, "
          f"stalls={result_no_fw['total_stalls']}, CPI={result_no_fw['cpi']}")
    print(f" With forwarding:      total_cycles={result_fw['total_cycles']}, "
          f"stalls={result_fw['total_stalls']} "
          f"(data={result_fw['data_hazard_stalls']}, control={result_fw['control_hazard_flushes']}), "
          f"CPI={result_fw['cpi']}")

    perf_no_fw = performance_metrics(result_no_fw["total_cycles"], clock_period_ns=1.0,
                                     n_instructions=len(program))
    perf_fw = performance_metrics(result_fw["total_cycles"], clock_period_ns=1.0,
                                  n_instructions=len(program))
    print(f"\n Performance @1GHz (1ns cycle):")
    print(f"  No forwarding : CPI={perf_no_fw['cpi']}, exec_time={perf_no_fw['execution_time_ns']}ns, "
          f"throughput={perf_no_fw['throughput_mips']} MIPS")
    print(f"  Forwarding   : CPI={perf_fw['cpi']}, exec_time={perf_fw['execution_time_ns']}ns, "
          f"throughput={perf_fw['throughput_mips']} MIPS")

    return {"no_forwarding": result_no_fw, "forwarding": result_fw,

```

```
"perf_no_forwarding": perf_no_fw, "perf_forwarding": perf_fw}
```

```
def demo_superscalar():  
    print("\n=== Superscalar / SMT Analytical Comparison (200-instruction benchmark) ===")  
    comp = superscalar_and_smt_comparison()  
    for label, data in comp.items():  
        print(f" {label:11s}: cycles={data['cycles']:>7} CPI={data['cpi']:.3f} IPC={data['ipc']:.3f}")  
    speedup_super = comp["scalar"]["cycles"] / comp["superscalar"]["cycles"]  
    speedup_smt = comp["scalar"]["cycles"] / comp["smt"]["cycles"] * 2 # SMT does 2x the work in fewer  
    cycles  
    print(f" Speedup, scalar -> superscalar: {speedup_super:.2f}x")  
    print(f" Effective speedup, scalar -> SMT (2 threads): {speedup_smt:.2f}x")  
    return comp
```

```
def main():  
    results = {}  
    results["alu"] = demo_alu()  
    results["cla"] = demo_cla()  
    results["booth"] = demo_booth()  
    results["division"] = demo_division()  
    results["ieee754"] = demo_ieee754()  
    results["pipeline"] = demo_pipeline()  
    results["superscalar"] = demo_superscalar()  
    print("\nAll functional units executed successfully.")  
    return results
```

```
if __name__ == "__main__":  
    main()
```